Rajkumar Arumugam Jeeva

rajkumaraj2303@gmail.com | 613-867-2638 | Toronto, ON

SUMMARY

AI Engineer with a solid Applied Science and Machine Learning background, specializing in building production-grade AI systems, LLM applications, and agentic workflows. Proven track record in engineering stateful multi-agent architectures (LangGraph), full-stack AI backends (FastAPI), and computer vision models. Combines robust software engineering principles with advanced machine learning to deploy scalable, high-impact intelligent systems.

PROFESSIONAL EXPERIENCE

AI Engineer | Branchy Solution | Toronto, ON June 2025 – Present

- Architected an autonomous project planning platform utilizing **LangGraph** to orchestrate complex multi-step agentic workflows, conditional routing, and stateful conversation management.
- Engineered typed state management using Pydantic and designed a persistent memory layer with **Supabase** to maintain cross-session states, enabling seamless, resumable, and long-running planning sessions.
- Developed a full-stack architecture combining a **FastAPI** backend and **React** frontend, leveraging real-time token streaming and structured LLM outputs to guide users from a raw idea to a finalized Project Blueprint.
- Established production-grade LLM observability via **LangSmith**, implementing automated prompt template management, execution tracing, and token debugging to optimize multi-agent workflow accuracy.

Research Associate | University of Ottawa | Ottawa, ON June 2025 – Present

- Developed and deployed scalable machine learning solutions on **Azure**, building automated production inference workflows and production-ready REST APIs via **FastAPI** and **Flask**.
- Monitored end-to-end model performance in production environments, conducting error analysis and mitigating data drift to ensure long-term model robustness.
- Collaborated closely with cross-functional data engineering and product stakeholders to transition advanced predictive analytics models into enterprise workflows.

Research Associate | Children's Hospital of Eastern Ontario (CHEO) Research Institute | Ottawa, ON Mar. 2024 – May 2025

- Built and optimized high-throughput data preprocessing pipelines for large-scale structured and unstructured datasets using **Pandas** and **NumPy**.
- Implemented supervised and unsupervised learning algorithms utilizing **Python** and **Scikit-learn**, significantly enhancing model metrics through rigorous feature engineering and hyperparameter tuning.
- Engineered data-driven predictive models for clinical analytics, validating system behavior through automated workflows and structured data pipelines.

ML Engineer | HTIC – Indian Institute of Technology Madras Research Park | India Aug. 2021 – Apr. 2023

- Engineered advanced computer vision and deep learning architectures (**CNNs**) for medical imaging and anomaly detection, accelerating diagnostic processing accuracy.
- Designed robust, end-to-end ML pipelines incorporating image augmentation, transfer learning, and multi-modal feature extraction across large-scale clinical datasets.
- Deployed optimized models onto **Azure** cloud environments to handle both real-time and batch inference workloads within integrated healthcare platforms.
- Conducted rigorous model evaluation, cross-validation, and production-level optimization to guarantee clinical-grade system reliability and performance robustness.

EDUCATION & SKILLS

Master of Applied Science in Biomedical Engineering | University of Ottawa | ON Sept. 2023 – June 2025
Bachelor of Engineering in Biomedical Engineering | Anna University | Chennai, India Aug. 2017 – May 2021

AI & Machine Learning: LLM Integration, Agent Orchestration (LangGraph), Prompt Engineering, Observability (LangSmith), Computer Vision, Deep Learning (CNNs)
Backend & Software Engineering: Python, SQL, FastAPI, Flask, RESTful API Design, Structured Outputs (Pydantic), C
Frontend & Databases: Next.js, React, Supabase, PostgreSQL, MongoDB, Data Preprocessing (Pandas, NumPy)
DevOps & Infrastructure: Azure (ML Workspace), Vercel, MLflow, Docker, Git, CI/CD Pipelines